Quantum Bill of Materials

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Readiness summary

Total crypto findings	40 across 20 distinct algorithms/protocols
Quantum-safe today (PQ_SAFE + HYBRID)	32.5%
Shor-vulnerable (existential break under a CRQC)	37.5%
Grover-weakened (margin reduction only)	5.0%
Already classically broken/deprecated	25.0%

Prioritised migration plan

#	Algorithm	Threat	Count	Exposure	Lifetime	Score	Replacement
1	ECC P-256 (ECDSA/ECDH) `EC-P256`	SHOR	3	internet_facing	15y	1205.1	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware
2	RSA (<2048-bit) `RSA-SUB2048`	SHOR	1	internet_facing	15y	1181.7	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware; and stop using <2048 now
3	MD5 `MD5`	CLASSICAL_BROKEN	3	internet_facing	15y	324.4	SHA-256 / SHA3-256 (or SHA-384 for CNSA 2.0)
4	ECB mode (misuse) `MODE-ECB`	CLASSICAL_BROKEN	2	internet_facing	15y	321.3	AES-256-GCM (AEAD)
5	X25519 `X25519`	SHOR	4	internal	7y	255.8	Hybrid X25519MLKEM768 now, ML-KEM-768 later
6	Ed25519 `ED25519`	SHOR	2	internal	7y	250.9	ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware
7	AES (key size unknown) `AES-UNK`	GROVER	2	internet_facing	15y	137.7	Verify key size; use AES-256
8	RSA (size unknown) `RSA-UNK`	SHOR	2	internal	1y	84.5	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware
9	DH/FFDH (<2048-bit) `DH-SUB2048`	SHOR	1	internal	1y	83.6	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first; stop using <2048 now

#	Algorithm	Threat	Count	Exposure	Lifetime	Score	Replacement
10	RSA-2048 RSA-2048	SHOR	1	internal	1y	83.6	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware
11	RSA-3072 RSA-3072	SHOR	1	internal	1y	83.6	ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware
12	SHA-1 SHA1	CLASSICAL_BROKEN	3	internal	2y	64.9	SHA-256 / SHA3-256 (or SHA-384 for CNSA 2.0)
13	3DES 3DES	CLASSICAL_BROKEN	1	internal	1y	63.6	AES-256-GCM
14	DES DES	CLASSICAL_BROKEN	1	internal	1y	63.6	AES-256-GCM
15	Hybrid KEM (classical+ML-KEM) HYBRID-KEM	HYBRID	3	internal	1y	9.3	Later: pure ML-KEM where policy allows
16	TLS 1.3 (hybrid PQ key exchange configured) TLS1.3-HYBRID	HYBRID	2	internal	1y	9.2	none needed -- verify the hybrid group is actually negotiated, not just configured
17	SHA-256 SHA-256	PQ_SAFE	3	internal	7y	6.2	none needed
18	ChaCha20(-Poly1305) CHACHA20	PQ_SAFE	2	internal	1y	3.7	none needed
19	ML-DSA (44/65/87) ML-DSA	PQ_SAFE	2	internal	1y	3.7	none needed
20	AES-256 AES-256	PQ_SAFE	1	internal	1y	3.6	none needed

Detail per item

1. ECC P-256 (ECDSA/ECDH) `(EC-P256)` — *score 1205.1*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 15y, against a ~10y conservative planning horizon.

Occurrences: 3

Affected components: payments-svc

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware

Locations: demo-run/workdir/certs/modern_service.crt:1, demo-run/workdir/certs/modern_service.key:1, demo-run/workdir/app/payments/service.py:11

2. RSA (<2048-bit) `(RSA-SUB2048)` — *score 1181.7*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 15y, against a ~10y conservative planning horizon.

Occurrences: 1

Affected components: payments-svc

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware; and stop using <2048 now

Locations: demo-run/workdir/app/payments/service.py:8

3. MD5 `(MD5)` — *score 324.4*

Already broken or deprecated against classical attackers today -- fix regardless of quantum timeline.

Occurrences: 3

Affected components: legacy-batch, payments-svc

Recommended replacement: SHA-256 / SHA3-256 (or SHA-384 for CNSA 2.0)

Locations: demo-run/workdir/web-service/crypto-utils.js:4, demo-run/workdir/app/payments/service.py:14, demo-run/workdir/app/legacy/hash.py:7

4. ECB mode (misuse) `(MODE-ECB)` — *score 321.3*

Already broken or deprecated against classical attackers today -- fix regardless of quantum timeline.

Occurrences: 2

Affected components: payments-svc

Recommended replacement: AES-256-GCM (AEAD)

Locations: demo-run/workdir/app/api/AuthService.java:17, demo-run/workdir/app/payments/service.py:17

5. X25519 `(X25519)` — *score 255.8*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 7y, against a ~10y conservative planning horizon.

Occurrences: 4

Affected components: vault-svc

Recommended replacement: Hybrid X25519MLKEM768 now, ML-KEM-768 later

Locations: demo-run/workdir/config/sshd_config:2, demo-run/workdir/config/nginx.conf:4, demo-run/workdir/config/openssl.cnf:6, demo-run/workdir/app/crypto/vault.py:6

6. Ed25519 (ED25519) — *score 250.9*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 7y, against a ~10y conservative planning horizon.

Occurrences: 2

Affected components: vault-svc

Recommended replacement: ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware

Locations: demo-run/workdir/config/sshd_config:5, demo-run/workdir/app/crypto/vault.py:9

7. AES (key size unknown) (AES-UNK) — *score 137.7*

Grover-weakened: effective security roughly halves under a large quantum computer; raise key/output size, no protocol redesign needed.

Occurrences: 2

Affected components: payments-svc, vault-svc

Recommended replacement: Verify key size; use AES-256

Locations: demo-run/workdir/app/crypto/vault.py:12, demo-run/workdir/app/payments/service.py:17

8. RSA (size unknown) (RSA-UNK) — *score 84.5*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 1y, against a ~10y conservative planning horizon.

Occurrences: 2

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware

Locations: demo-run/workdir/config/sshd_config:5, demo-run/workdir/app/api/AuthService.java:6

9. DH/FFDH (<2048-bit) (DH-SUB2048) — *score 83.6*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 1y, against a ~10y conservative planning horizon.

Occurrences: 1

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first; stop using <2048 now

Locations: demo-run/workdir/config/sshd_config:2

10. RSA-2048 (RSA-2048) — *score 83.6*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 1y, against a ~10y conservative planning horizon.

Occurrences: 1

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware

Locations: demo-run/workdir/app/api/AuthService.java:7

11. RSA-3072 (RSA-3072) — *score 83.6*

Shor-breakable: an attacker recording traffic/data today can decrypt it once a CRQC exists. Data here must stay confidential 1y, against a ~10y conservative planning horizon.

Occurrences: 1

Recommended replacement: ML-KEM-768 (FIPS 203); in TLS/SSH use hybrid X25519MLKEM768 first / ML-DSA-65 (FIPS 204); SLH-DSA (FIPS 205) for long-lived roots/firmware

Locations: demo-run/workdir/config/openssl.cnf:2

12. SHA-1 (SHA1) — *score 64.9*

Already broken or deprecated against classical attackers today -- fix regardless of quantum timeline.

Occurrences: 3

Affected components: legacy-batch

Recommended replacement: SHA-256 / SHA3-256 (or SHA-384 for CNSA 2.0)

Locations: demo-run/workdir/config/sshd_config:4, demo-run/workdir/app/api/AuthService.java:12, demo-run/workdir/app/legacy/hash.py:4

13. 3DES (3DES) — *score 63.6*

Already broken or deprecated against classical attackers today -- fix regardless of quantum timeline.

Occurrences: 1

Recommended replacement: AES-256-GCM

Locations: demo-run/workdir/web-service/crypto-utils.js:12

14. DES (DES) — *score 63.6*

Already broken or deprecated against classical attackers today -- fix regardless of quantum timeline.

Occurrences: 1

Recommended replacement: AES-256-GCM

Locations: demo-run/workdir/app/api/AuthService.java:17

15. Hybrid KEM (classical+ML-KEM) (HYBRID-KEM) — *score 9.3*

Already hybrid classical+PQ; low urgency, monitor for pure-PQ migration once policy allows.

Occurrences: 3

Recommended replacement: Later: pure ML-KEM where policy allows

Locations: demo-run/workdir/config/sshd_config:2, demo-run/workdir/config/nginx.conf:4, demo-run/workdir/config/openssl.cnf:6

16. TLS 1.3 (hybrid PQ key exchange configured) (TLS1.3-HYBRID) — *score 9.2*

Already hybrid classical+PQ; low urgency, monitor for pure-PQ migration once policy allows.

Occurrences: 2

Recommended replacement: none needed -- verify the hybrid group is actually negotiated, not just configured

Locations: demo-run/workdir/config/nginx.conf:3, demo-run/workdir/config/openssl.cnf:5

17. SHA-256 (SHA-256) — *score 6.2*

Already quantum-safe with adequate margin.

Occurrences: 3

Affected components: vault-svc

Recommended replacement: none needed

Locations: demo-run/workdir/web-service/crypto-utils.js:8, demo-run/workdir/certs/modern_service.crt:1, demo-run/workdir/app/crypto/vault.py:17

18. ChaCha20(-Poly1305) (CHACHA20) — *score 3.7*

Already quantum-safe with adequate margin.

Occurrences: 2

Recommended replacement: none needed

Locations: demo-run/workdir/config/sshd_config:3, demo-run/workdir/config/nginx.conf:5

19. ML-DSA (44/65/87) (ML-DSA) — *score 3.7*

Already quantum-safe with adequate margin.

Occurrences: 2

Recommended replacement: none needed

Locations: demo-run/workdir/certs/legacy_partner.crt:1, demo-run/workdir/certs/legacy_partner.key:1

20. AES-256 (AES-256) — *score 3.6*

Already quantum-safe with adequate margin.

Occurrences: 1

Recommended replacement: none needed

Locations: demo-run/workdir/config/sshd_config:3